



Sri Chaitanya
Educational Institutions



JEE ADVANCED GRAND TEST



Sri Chaitanya IIT Academy.,India.

✦ A.P ✦ T.S ✦ KARNATAKA ✦ TAMILNADU ✦ MAHARASTRA ✦ DELHI ✦ RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec: **Sr.Super60_Elite, Target & LIIT-BTs**

Paper -2(Adv-2022-P2-Model)

Date: 12-05-2024

Time: 02.30Pm to 05.30Pm

GTA-32

Max. Marks: 180

12-05-2024_**Sr.Super60_Elite, Target & LIIT-BTs**_Jee-Adv(2022-P2)_GTA-32_Syllabus

MATHEMATICS : TOTAL SYLLABUS

PHYSICS : TOTAL SYLLABUS

CHEMISTRY : TOTAL SYLLABUS

Name of the Student: _____

H.T. NO:

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**JEE-ADVANCE-2022-P2-Model**

Time:3Hr's

IMPORTANT INSTRUCTIONS

Max Marks: 180

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 8)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 9 – 14)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 15 – 18)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 19 – 26)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 27 – 32)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 33 – 36)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 37 – 44)	Questions with Integer Answer Type	+3	-1	8	24
Sec – II(Q.N : 45 – 50)	Questions with Multiple Correct Choice with partial mark	+4	-2	6	24
Sec – III(Q.N : 51 – 54)	Questions with Single Correct Choice	+3	-1	4	12
Total				18	60

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Space for rough work

Page 2

**Sri Chaitanya**
Educational Institutions**Infinity**
Learn**THE PERFECT HAT-TRICK WITH ALL- INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**JEE MAIN
2023
SINGARAJU
VENKAT KOUNDINNYA
Sri Chaitanya
Elite Class
300
300**RANK**
1JEE Advanced
2023
VARUN
CHIDVILAS REDDY
Sri Chaitanya
Elite Class
341
360**RANK**
1NEET
2023
BODA VARUN
CHAKRAVARTHI
Sri Chaitanya
Elite Class
720
720**RANK**
1



MATHEMATICS

Max Marks: 60

SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
 Full Marks: +3 If ONLY the correct integer is entered;
 Zero Marks: 0 If the question is unanswered;
 Negative Marks: -1 In all other cases

1. Let A be 3×3 involuntary symmetric matrix, such that $X = APA'$ where

$$P = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \text{ and } |A| \neq 0, \text{ then } \text{Tr}(A'X^{2023}A^{-1} + I) \text{ is}$$

2. Two roots of the equation $x^4 - 6x^3 + 18x^2 - 30x + 25 = 0$ are of the form

$$\alpha + i\beta, \beta + i\alpha, \text{ where } \alpha, \beta \in \mathbb{R}. \text{ Then } \alpha^2 + \beta^2 =$$

3. Let $a_1, a_2, a_3, \dots, a_n$ be a sequence of numbers satisfying

$$(n+1)(a_n - a_{n+1}) = \left(\frac{n^2 + 3n + 2}{3^n} \right) - a_n \quad \forall n \in \mathbb{N}. \text{ If } a_1 = 7, \text{ then } \lim_{n \rightarrow \infty} \left(\frac{a_n}{n+1} \right) \text{ is}$$

4. Let $\vec{a} = \frac{1}{7}(2\hat{i} + 3\hat{j} + 6\hat{k})$; $\vec{b} = \frac{1}{7}(6\hat{i} + 2\hat{j} - 3\hat{k})$ and $\hat{c}$ be a unit vector and

$$A = \begin{bmatrix} \frac{2}{7} & \frac{3}{7} & \frac{6}{7} \\ \frac{6}{7} & \frac{2}{7} & -\frac{3}{7} \\ \hat{c} \cdot \hat{i} & \hat{c} \cdot \hat{j} & \hat{c} \cdot \hat{k} \end{bmatrix} \text{ and , then } \left[3|\hat{c} \cdot \hat{i}| + 2|\hat{c} \cdot \hat{j}| + |\hat{c} \cdot \hat{k}| \right] \text{ is (where } [\cdot] \text{ represents greatest}$$

integer function)

5. Given that set S has four odd integers and their range is 4, how many distinct values can the standard deviation of S take?





6. Let A and B be two independent events. If $a, b \in R$ such that

$$a \left(\sqrt{P\left(\frac{A}{B}\right)} \right) + b \left(\sqrt{P\left(\frac{\bar{A}}{B}\right)} \right) = \frac{2}{3}. \text{ If least value of } a^2 + b^2 = k, \text{ then } \left\lceil \frac{1}{k} \right\rceil \text{ is (where } \lceil \cdot \rceil \text{ represents greatest integer function)}$$

7. If recursion polynomials $P_k(x)$ are defined as

$$P_1(x) = (x-2)^2, P_2(x) = \left((x-2)^2 - 2 \right)^2$$

$$P_3(x) = \left((x-2)^2 - 2 \right)^2 - 2 \dots \dots \dots \text{In general}$$

$$P_k(x) = \left(P_{k-1}(x) - 2 \right)^2 \text{ then, find the constant term in } P_k(x).$$

8. If nine numbers $1, 2, 3, 4, 5, 6, 7, 8, 9$ are put into a matrix of order 3×3 randomly so that each number occur exactly once. If the probability that the sum of the numbers in atleast one row is greater than 21 is $\frac{1}{p}$, then the value of p is

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.

•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

•For each question, choose the option(s) corresponding to (all) the correct answer(s).

•Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks** +3 If all the four options are correct but **ONLY** three options are chosen;

Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

9. Let $a_i \in R - \{0\}$ for $i = 1, 2, 3$ such that they are all distinct. Let $f(x)$, $g(x)$ and $h(x)$ be real valued functions given by

$$h(x) = xg(x) + 2f(x) - x$$

$$g(x) = \frac{x^2 + a_1^2 + 2a_1x}{(a_1 - a_2)(a_1 - a_3)} + \frac{x^2 + a_2^2 + 2a_2x}{(a_2 - a_3)(a_2 - a_1)} + \frac{x^2 + a_3^2 + 2a_3x}{(a_3 - a_1)(a_3 - a_2)},$$

$$f(x) = \begin{vmatrix} 0 & a_1 - x & a_2 - x \\ -a_1 - x & 0 & a_3 - x \\ -a_2 - x & -a_3 - x & 0 \end{vmatrix} (a_i \neq 0)$$

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Space for rough work

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**THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023**

**JEE MAIN
2023**
SINGARAJ
VENKAT KOUNDUR
RANK 1
300
300



**RANK
1**

**JEE Advanced
2023**
SIVAKUMAR
CHIDVILAS REDDY
RANK 1
341
360



**RANK
1**

**NEET
2023**
SODA VARUN
CHAKRAVARTI
RANK 1
720
720



**RANK
1**



Which of the following statements is/are CORRECT?

A) $h(x) = 0$ has three distinct real roots

B) If $\frac{1}{a_2} = \frac{1}{a_1} + \frac{1}{a_3}$, then $h(x) = 0$ has three distinct real roots

C) If $\frac{2}{a_2} = \frac{1}{a_1} + \frac{1}{a_3}$, then $h(x) = 0$ has three distinct real roots

D) Product of all roots of $h(x) = 0$

10. The equation of plane equidistant from lines $\frac{x-1}{1} = \frac{y-2}{2} = \frac{z-3}{3}$ and

$2x - y - 4 = 0 = 3x - z - 8$ is/are

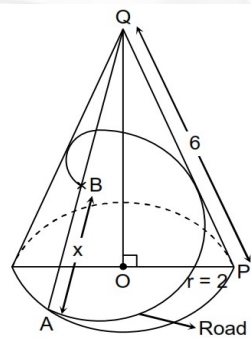
A) $x + y - z - 2 = 0$

B) $x - 2y + z = 5$

C) $(2 + 3\lambda)x - y - \lambda z - 2(2 + \lambda) = 0 \quad \forall \lambda \in \mathbb{R}$

D) $2x - 4y + 2z = 3p \quad \forall p \in \mathbb{R}$

11. The figure illustrates a right circular cone shaped mountain. If a shortest distance road is built for sight-seeing around the mountain, in which road starts at point A and ends at B (A, B and Q are collinear). Let $f(x)$ be the total length of road. The road will go up-hill for some distance $L(x)$ and down-hill for the distance of $g(x)$.



Given $PQ = 6$; $OP = 2$; $AB = x$, then

A) $f(x) = \sqrt{x^2 - 18x + 108} \quad \forall x \in [0, 6]$

B) Maximum value of $g(x)$ is $3\sqrt{3}$

C) $g(x) = \frac{f(x)}{\sqrt{x^2 - 15x + 54}} \quad \forall x \in [0, 6]$

D) If at $x = x_0$, $g(x)$ take a maximum value, then $f(x_0) = 6\sqrt{3}$





12. $f(x)$ is a continuous and increasing function satisfying

$$f(x) = f(x-3) + 4 \text{ and } \int_0^6 f(x) dx = 0. \text{ Then the CORRECT option(s) is/are}$$

- A) Area bounded by $y = f(x)$, x -axis and lines $x = 6$ and $x = 9$ is 9
 B) Area bounded by $y = f(x)$, x -axis and lines $x = 6$ and $x = 9$ is 18
 C) $y = f(x)$ is a periodic function
 D) if $y = f(x) + h(x)$ is periodic function (where $h(x)$ is continuous function), then

$$\lim_{x \rightarrow \infty} \frac{(h(x) - x)(h(x) + x)}{x^2} = \frac{7}{9}$$
13. ABCD is a tetrahedron with $\triangle BCD$ being equilateral, having each side equal to 12 units. H is foot of perpendicular from A to $\triangle BCD$ and it lies inside the triangle if area of $\triangle CDH$ is three times the area of $\triangle BCH$ and area of $\triangle DBH$ is two times the area of $\triangle BCH$, $AH = 3$ and M is midpoint of side BD. Choose the correct option(s)
 A) perpendicular distance of point H from side BC is $\sqrt{3}$
 B) perpendicular distance of point H from side BD is $2\sqrt{3}$
 C) perpendicular distance of point A from line CM is $\sqrt{13}$
 D) perpendicular distance of point A from line CM is $\sqrt{15}$
14. For a given parabola $y^2 = 4ax$, two variable chords PQ and RS at right angles are drawn through the fixed point $A(x_1, y_1)$ inside the parabola, making variable angles θ and α with x -axis. If r_1, r_2, r_3, r_4 are distances of P, Q, R and S from A , then the value of $\frac{1}{r_1 r_2} + \frac{1}{r_3 r_4}$
 A) independent of θ
 B) independent of α
 C) depends upon both θ & α
 D) is a constant

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

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Space for rough work

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THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1
IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

JEE MAIN 2023
RANK 1
 SINGARAJU VENKAT KOUNDUR
 Sri Chaitanya
 Elite 2m Class
300
300



JEE Advanced 2023
RANK 1
 VADIVELA CHIDVILAS REDDY
 Sri Chaitanya
 Elite 12m Class
341
360



NEET 2023
RANK 1
 BORA VARUN CHAKRAVARTHI
 Sri Chaitanya
 Elite 2m Class
720
720





- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

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Negative Marks : -1 In all other cases

15. If $\alpha \in \mathbb{R}$ be such that the point $P(\alpha, 4 - \alpha^2)$ lies inside the triangle equation of whose three sides are $y - tx - t - 3 = 0$; $ty + x - 2 = 0$ and $y = 5$, where t is a parameter and $t \in (-\infty, -3]$, then

A) $\alpha \in (0, 3)$

B) $\alpha \in (-\infty, t) \forall t \in (-\alpha, -3]$

C) $\alpha \in (-1, 2)$

D) $\alpha \in (t, \alpha) \forall t \in (-\alpha, -3]$

16. If $A > 0, c, d, u, v$ are non zero constants, and the graphs of $f(x) = |Ax + c| + d$ and $g(x) = -|Ax + u| + v$ intersect exactly at two points $(1, 4)$ and $(3, 1)$, then the value of $\frac{u+c}{A}$ equals:

A) 4

B) -4

C) 2

D) -2

17. Let $f: [0, 4] \rightarrow \mathbb{R}$ is continuous function such that $|f(x)| = 6|(x - n + 1)(x - n)|$ where

$n \in \mathbb{N} \forall x \in [n-1, n]$ and $g(x) = \int_0^x f(t) dt - \int_x^4 f(t) dt$ has maximum at $x = 2$, then

$\int_{1/2}^4 f(t) dt$ is/are

A) $-\frac{1}{2}$

B) 0

C) $\frac{1}{2}$

D) 1

18. If $\sum_{k=0}^9 (-1)^k 3^{9-k} {}^{20}C_k {}^{20-k}C_{9-k} 2^{-9} = {}^{20}C_r$, then value of r is

A) 8

B) 10

C) 11

D) 13



PHYSICS

Max Marks: 60

SECTION-I
(INTEGER ANSWER TYPE)

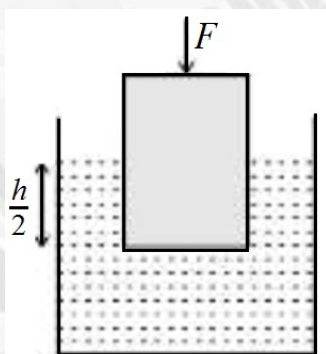
- This section contains EIGHT (08) questions.
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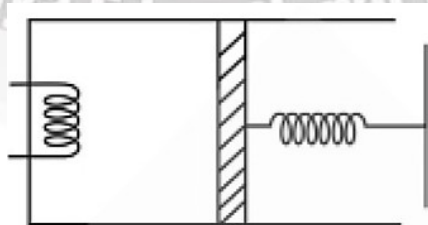
Zero Marks: 0 If the question is unanswered;

Negative Marks: -1 In all other cases

19. A liquid of density ρ is filled in a cylindrical vessel whose base radius is R . A wooden cylinder of height h , base radius $R/2$ and density $\rho/2$ is floating in the liquid. Minimum work to be done by external agent in completely immersing the cylinder into liquid is $\frac{n\pi R^2 \rho g h^2}{128}$. Find the value of n .



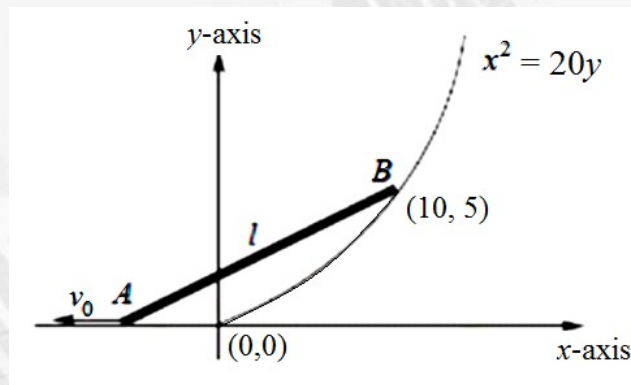
20. An ideal monoatomic gas is confined by a spring loaded massless piston of cross-section $8 \times 10^{-3} \text{ m}^2$. Initially the gas is at 300 K and occupies a volume of $2.4 \times 10^{-3} \text{ m}^3$ and the spring is in its relaxed state. The gas is heated by an electric heater until piston moves out without friction by 0.1 m. Force constant of the spring is 8000 N/m, atmospheric pressure $1 \times 10^5 \text{ N/m}^2$, the cylinder and piston are thermally insulated, the piston and spring are massless and there is no friction between the piston and cylinder. Neglect the heat lost through the wire of the heater. The heat capacity of the heater is neglected.



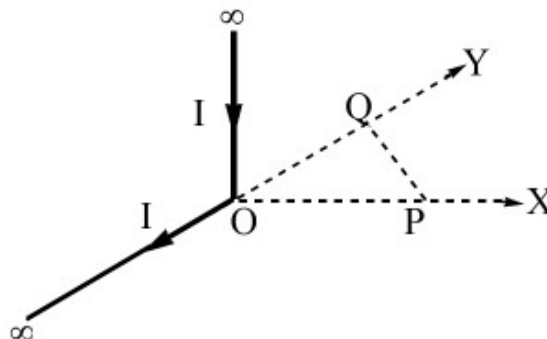
Find the ratio of heat supplied by the heater to the work done by the gas



21. A moving neutron (of mass m) is about to collide with a stationary singly ionized helium atom (of mass $4m$) in ground state. The speed of the incident neutron is equal to the minimum speed required such that the collision may be inelastic. If the collision is head-on perfectly elastic then the energy of neutron after collision is $\frac{n}{5}(10.2eV)$ where n is a positive integer. Find n ? (Assume Bohr's quantization principle to be valid for energy states of singly ionized helium atom and $1 \text{ Rydberg} = 13.6eV$)
22. A uniform rod has mass $m = 2kg$ and length $l = 13m$. One end of the rod is pulled with a constant velocity of $v_0 = 34m/s$ along a frictionless horizontal floor in the negative x direction. The other end is moving along a parabolic fixed curve. The equation of the parabola is $x^2 = 20y$. Find the angular velocity of the rod (in rad/s) when the end point 'B' is at $(10, 5)$

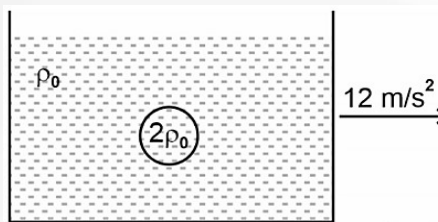


23. Figures Shows combination of two semi infinite long wires kept on Z and Y-axis. It carries constant current I as shown in figure. $\int_Q^P \vec{B} \cdot d\vec{l}$ on line QP is $x\mu_0 i$. Then fill the value of $\frac{1}{x}$ _____. Co-ordinate of point P and Q are $(a, 0, 0)$ and $(0, a, 0)$ respectively

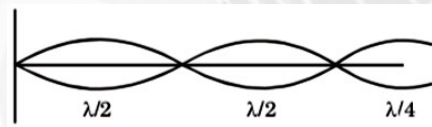




24. A vessel containing liquid of density ρ_0 is accelerated horizontally with constant acceleration 12 m/s^2 in gravity free space. A small sphere of density $2\rho_0$ is released from rest with respect to the vessel. Magnitude of initial horizontal acceleration of sphere with respect to vessel is..... m/s^2



25. A string of length ℓ fixed at one end and free at another end is vibrating in 5^{th} harmonic. Maximum displacement of particle is A . Amplitude of oscillation of a particle at distance $\frac{\ell}{10}$ from free end is $\frac{A}{\sqrt{\alpha}}$ find value of α .



26. A certain amount of mono atomic gas undergoes a process for which $U^\beta \alpha V$, where U is internal energy and V is volume of gas. It is found that the ratio $\frac{\Delta U}{\Delta Q}$ for the process is $\frac{1}{3}$. What is the value of β .

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.

•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

•For each question, choose the option(s) corresponding to (all) the correct answer(s).

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Partial Marks: +1 If two or more options are correct but **ONLY** one option is chosen and it is a correct option;

Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.

27. A source, approaching with speed u towards the open end of a stationary pipe of length L , is emitting a sound of frequency f_s . The farther end of the pipe is closed. The speed of sound in air is v and f_0 is the fundamental frequency of the pipe. For which of the following combination(s) of u and f_s , will the sound reaching the pipe lead to a resonance?

A) $u = 0.8v$ and $f_s = f_0$

B) $u = 0.8v$ and $f_s = 2f_0$

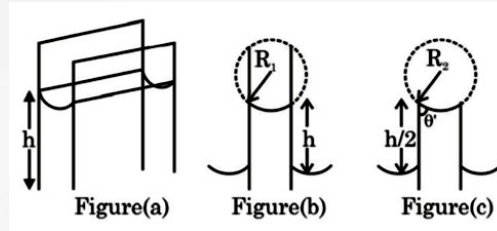
C) $u = 0.8v$ and $f_s = 0.5f_0$

D) $u = 0.5v$ and $f_s = 1.5f_0$





28. Two parallel plates at a distance d are submerged in water. (Surface tension = s and contact angle θ) Water has density ρ . The figure (a) shows the water rise and cross section view is in figure (b). Now the plates are pushed down so that the height of plates above water is $\frac{h}{2}$ as in figure (c).

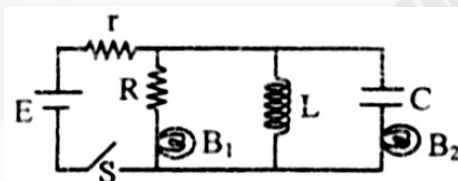


- A) $h = \frac{2s \cos \theta}{\rho dg}$ B) $\cos \theta' = \frac{\cos \theta}{2}$ C) $R_1 = \frac{d}{2 \cos \theta}$ D) $R_2 = \frac{d}{4 \cos \theta}$
29. A positive point charge ' q ' of mass ' m ' kept at a distance x_0 (in the same plane) from a fixed very long straight wire carrying current ' I ' is projected normally away from it with speed ' v ' (in the same plane). Which of the following statements is correct for subsequent motion (ignore gravity):
- A) The velocity of the particle remains constant
 B) The angular momentum of the particle along the wire does not change as the charge moves
 C) The maximum separation of the particle from the wire is given by $x = x_0 e^{2pmv/m_0qi}$
 D) The radius of curvature of the path of particle remains constant
30. A particle moves on a circular path of radius R , so that its angular velocity about a fixed point on the circumference of the circular path is constant and is equal to ω . Then
- A) the acceleration of the particle is variable.
 B) the acceleration of the particle is constant in magnitude and is equal to $4R\omega^2$
 C) the acceleration of the particle varies in magnitude only
 D) acceleration of the particle will be zero.





31. The potential energy ' U ' in Joule of a particle of mass 1 kg, moving in $X - Y$ plane, obeys the law $U = 3x + 4y$, where (x, y) are the co-ordinates of the particle in meter. If the particle is at rest at $(6, 4)$ at $t = 0$, then (Assuming there are no non-conservative forces acting on the particle)
- A) the particle has constant acceleration
 B) the work done by the external forces, from the position of rest of the particle to the instant of crossing the X-axis is 50 J
 C) the speed of the particle when it crosses the Y-axis is 10 ms^{-1}
 D) the co-ordinates of the particle at time $t = 4 \text{ s}$ are $(-18, -28)$
32. In the given circuit at $t = 0$, switch S is closed, B_1 and B_2 having resistance r_1 ($r_1 < R$)



- A) At $t = 0$, B_2 will glow brighter
 B) At $t = \infty$, B_1 will glow brighter
 C) At $t = \infty$, B_2 will stop glowing
 D) At $t = \infty$, B_1 will stop glowing

SECTION - III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

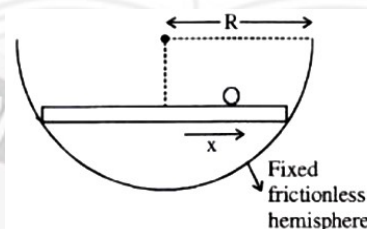
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Negative Marks : -1 In all other cases

33. A uniform rod of mass M and length L ($< 2R$) is placed symmetrically inside a fixed hemispherical surface (see figure). An insect of mass m is moving on rod such that rod remains in equilibrium in horizontal position (x is instantaneous displacement of insect from centre of rod). Then



- A) situation is not possible
 B) situation is possible if acceleration of insect is proportional to x
 C) situation is possible if there is no friction between rod and insect
 D) situation is possible if M/m is above certain minimum value

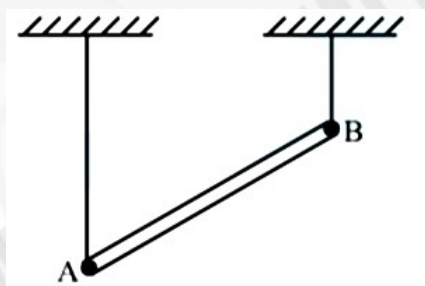




34. A body of mass m with specific heat C at temperature T_h is brought into thermal equilibrium by contact with an identical body which was initially at a lower temperature T_l . No heat is lost to the surroundings and the temperatures are in kelvin. The change in entropy of the system is

A) $mC \ln \left(\frac{(T_h + T_l)^2}{4T_l T_h} \right)$ B) $mC \ln \left(\frac{(T_h - T_l)^2}{4T_l T_h} \right)$
 C) $mC \ln \left(\frac{T_h}{T_l} \right)$ D) 0

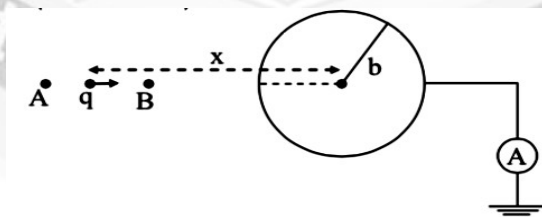
35. Two strings support a uniform rod as shown. String at end B is cut. Which of the following is true just after cut?



- [P] initial acceleration of A is vertical
 [Q] initial acceleration of A is horizontal
 [R] initial acceleration of centre of mass of rod is vertical
 [S] initial acceleration of centre of mass of rod is horizontal

A) P and Q B) Q and R C) R and S D) P and S

36. If a charge q is moving towards the centre of an earthed conducting sphere of radius b with uniform velocity v . Distance of two points A and B from centre of sphere are $3a$ and $2a$. Conducting sphere is earthed with an ammeter and resistance in series (as shown).



Reading a ammeter at this instant

A) $q \frac{xV}{b^2}$ B) $q \frac{V}{b}$ C) $q \frac{bV}{x^2}$ D) Zero



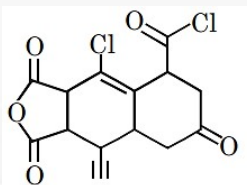
CHEMISTRY

Max Marks: 60

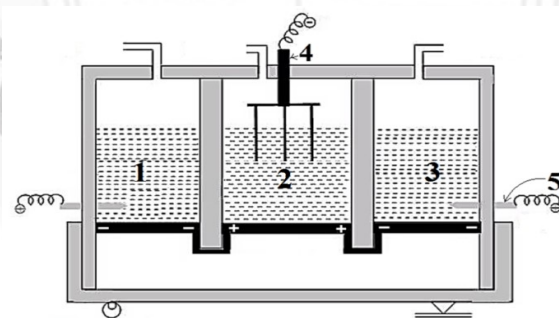
SECTION-I
(INTEGER ANSWER TYPE)

- This section contains EIGHT (08) questions.
- The answer to each question is a SINGLE DIGIT INTEGER ranging from 0 TO 9, BOTH INCLUSIVE.
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual Numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:
 Full Marks: +3 If ONLY the correct integer is entered;
 Zero Marks: 0 If the question is unanswered;
 Negative Marks: -1 In all other cases

37. Find out total number of moles of Grignard reagent $RMgX$ consumed when treated it with 1 mole of 'X'



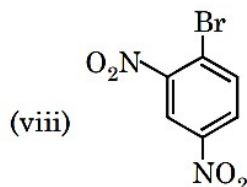
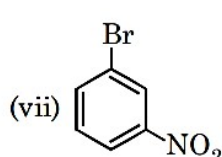
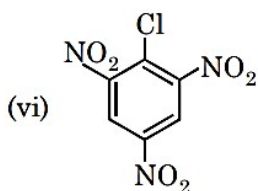
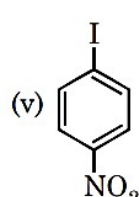
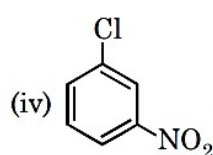
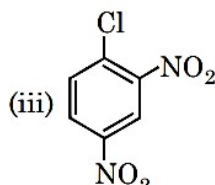
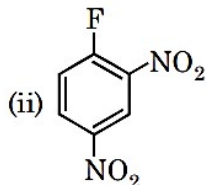
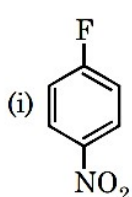
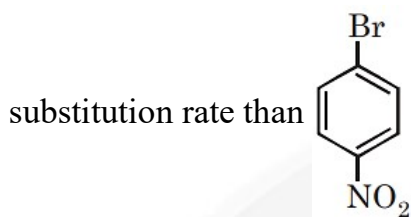
38. If the number of C-atoms in **Indane** are m and the number of H-atoms in **Lindane** are n , report the value of $m - n$.
39. Find the number of paramagnetic compound(s) with octahedral anion.
 $K_2[CoF_6]$; $O_2[AsF_6]$; $K_2[CoCl_4]$; $K_3[Co(CN)_6]$;
 $K_3[CoF_6]$; $[Ni(NH_3)_6]SO_4$; $[Pt(NH_3)_6][Fe(CN)_6]$; $Na_2[Fe(CN)_5NO]^{II}$
40. When 1 mole H_3PO_3 is neutralized with excess NaOH, ΔH is -106.68 kJ . If the enthalpy of neutralization of 1 mole HCl with excess NaOH is -55.84 kJ , calculate $\Delta H_{\text{ionisation}}$ of H_3PO_3 in kJ/mole .
41. Take a careful look at the figure of Castner Kellner cell used for sodium hydroxide production.



The sum of chamber/chambers containing brine solution and the number associated with graphite anode is _____



42. How many of the following compounds given below have faster aromatic nucleophilic



43. A current of $80\mu\text{A}$ is passed through AgNO_3 solution for 96.5 minutes using Pt electrodes. 75% of the cathode is occupied by a single atom thick silver layer. If one silver atom occupies a surface area of $1.5625 \times 10^{-16} \text{ cm}^2$, then the total surface area of the cathode is $z \times 10^2 \text{ cm}^2$. Given $N_A = 6 \times 10^{23}$, the value of z is _____
44. The formation constant of $[\text{Cu}(\text{NH}_3)_4]^{+2}$ is 1.25×10^{12} . When 0.0125 moles of copper is completely oxidised to +2 oxidation state and put into 1L of 0.25M NH_3 solution, the equilibrium concentration of the non-complexed cupric ion is found to be $a.bc \times 10^d$ (a point bc into ten raised to the power d). Calculate the value of $a + b + c + d$

SECTION – II (ONE OR MORE CORRECT ANSWER TYPE)

•This section contains SIX (06) questions.

•Each question has FOUR options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

•For each question, choose the option(s) corresponding to (all) the correct answer(s).

•Answer to each question will be evaluated **according to the following marking scheme:**

Full Marks: +4 If only (all) the correct option(s) is(are) chosen; **Partial Marks** +3 If all the four options are correct but ONLY three options are chosen;

Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct;

Partial Marks: +1 If two or more options are correct but ONLY one option is chosen and it is a correct option;

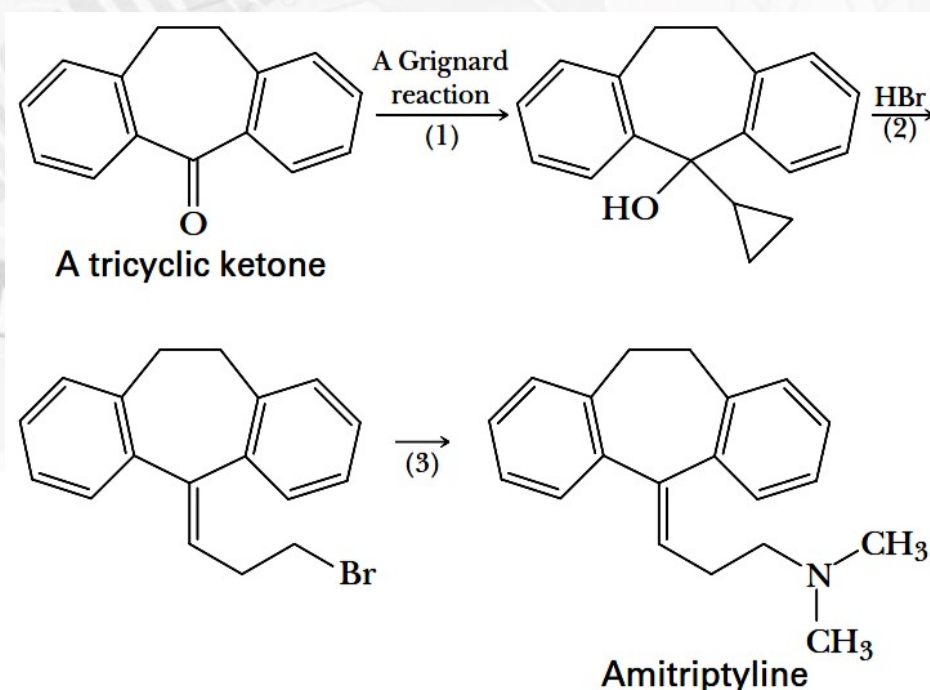
Zero Marks: 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks: -2 In all other cases.





45. At 300 K, a 0.5 mole sample of air (80% by mol of N_2 & 20% by mol of O_2) is put in 10 litre cylinder piston system in which 5 litre water is already present, then
- k_H for $N_2 = 10^5$ atm
- k_H for $O_2 = 4 \times 10^4$ atm
- A) X_{N_2} in H_2O will be 1.9704×10^{-5}
- B) X_{O_2} in H_2O will be 1.2315×10^{-5}
- C) On increasing temperature value of Henry's constant increases
- D) In H_2O , $N_2(g)$ is more soluble as compare to $O_2(g)$ at same pressure
46. Which of the following processes is/are involved in the extraction of Cu from sulphide ores of copper containing iron sulphide.
- A) FeS is more readily converted to the oxide than Cu_2S
- B) Liq. Matte is placed in converter for carbon reduction
- C) Cu_2S is partially converted to Cu_2O by air blasting and then to metallic copper
- D) Silica is added to combine with FeO
47. Amitriptyline, sold under the brand name **Elavil** is a tricyclic antidepressant. The flowchart below outlines its chemical synthesis.

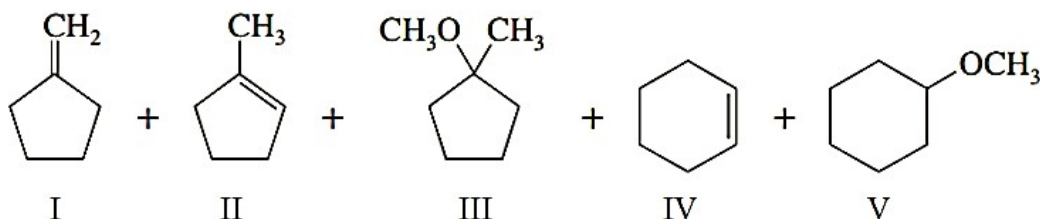


Select the correct statement(s) :





- A) Δ MgI is a possible Grignard reagent in step-1 followed by acid workup
 B) Step-2 involves the formation of a highly stable carbocation intermediate
 C) a primary carbocation is not formed as an intermediate at any point during the drug making
 D) diethyl amine attacks in an S_N2 manner in step-3
48. Given : $H_2O_2 + KMnO_4 + H_2SO_4 \rightarrow MnSO_4 + H_2O + O_2 + K_2SO_4$
 which is/are correct statement(s)?
 A) Number of gm equivalent of H_2O_2 and $KMnO_4$ reacts in the ratio of 5 : 2
 B) Number of moles of $MnSO_4$ and O_2 produced in the ratio of 5 : 2
 C) Equivalent mass of H_2SO_4 is 29.4
 D) 100 ml 2.27 V H_2O_2 reacts completely with 160ml 0.5M $KMnO_4$ in given reaction
49. Solvolysis of bromomethylcyclopentane in methanol gives a complex product mixture of the following 5 organic compounds



Identify the false statement(s)

- A) compound-II is formed by E_2 mechanism
 B) compound-III is formed by S_N2 mechanism
 C) in the formation of compounds I & IV, methanol acts as a base
 D) in the formation of compound V, methanol acts as a nucleophile
50. The following first order reaction –
 $2 A(aq.) \rightarrow 3 B(aq.) + 4C(aq.)$
 is monitored by measuring optical rotation of reaction mixture containing optical active dextrorotatory inert impurity at different time interval. The species A, B and C are optically active with specific rotation 20° , 40° and -80° per mol respectively.

Time (min)	0	22	∞
Optical rotation of reaction mixture	120°	-40°	-120°





(Given $-\ln 3 = 1.1, \ln 2 = 0.7$)

which is/are correct statements :

- A) Rate constant of reaction is 0.05 min^{-1}
- B) Rate constant of appearance of C is 0.1 min^{-1}
- C) The time in which 75% of A reacted is 28 min
- D) The time in which 50% of A reacted is 420 sec.

SECTION – III
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;

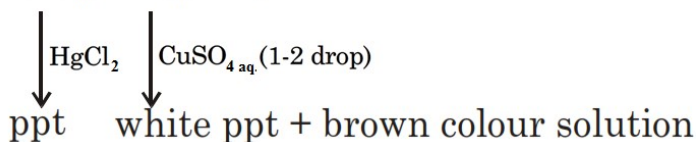
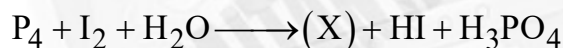
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);

Negative Marks : -1 In all other cases

51. Which is **INCORRECT** statement?

- A) Brownian movement is responsible for the stability of colloidal solution
- B) No Tyndall effect observed in True solution
- C) Haemoglobin (blood) and gold sol are negative charge colloids
- D) Casein act as emulsifier for milk

52. Select **INCORRECT** for given reaction sequence.

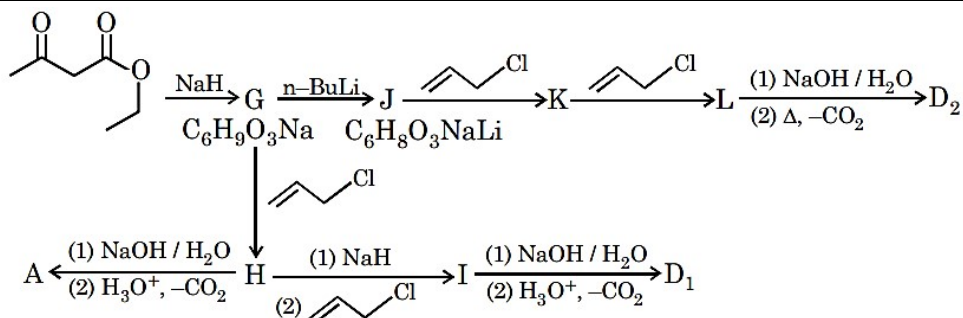


- A) $\text{Y}_{(\text{g})}$ produce ppt with CuSO_4
- B) Reaction of $\text{Z}_{\text{aq.}} + \text{CuSO}_4$ is redox change
- C) Bond angle in $\text{P}_4 < (\text{Y})_{(\text{g})}$
- D) On heating H_3PO_4 it produce $\text{Y}_{(\text{g})}$

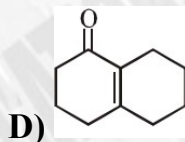
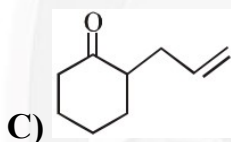
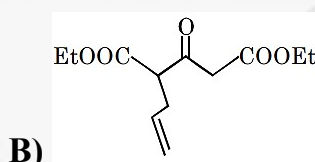
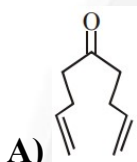
Paragraph for Questions 53 and 54

One of the best methods for the regioselective alkylation of ketones consists in an activation of the required α – carbon atom. For example :

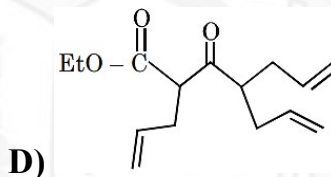
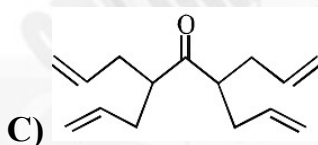
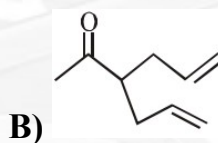
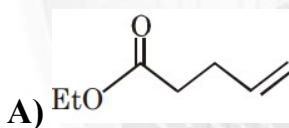




53. Identify structure of 'D₂' ?



54. Identify structure of D₁ ?





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